Baker College Teacher Prep Lesson Plan Format

Subject Area & Grade Level:

Lesson Duration:

Lesson Goal: What do we want students to learn?	Assessment: How will we know if they have learned it? (see guide)	Intervention: What will we do if they don't learn it? (see guide)	Enrichment: What will we do if they already know it? (see guide)
Students will apply and extend previous understandings of multiplication and division to divide fractions by fractions. They will interpret and compute quotients of fractions, represent problems with visual fraction models and equations, and connect division to multiplication using the general rule (a/b) ÷ (c/d) = ad/bc. As the Michigan Department of Education (n.d.) explains on page 44, students must be able to solve word problems involving division of fractions by fractions using visual fraction models and equations to represent the problem.	As Dance and Kaplan (2018) suggest from Chapter 5, strategic, deliberate questions can push student thinking to deeper levels and open up opportunities for quick formative assessment. Thus, assessment is embedded throughout the lesson via the following: • ☑ Diagnostic is a I Notice / I Wonder launch to check prior knowledge of fraction operations • ☑ Formative is a teacher observation and strategic questioning during guided practice and cooperative group work; student self-assessment (thumbs up/sideways/down) ☑ Summative is an exit ticket were students independently solve one fraction word problem from the standard and write a sentence justifying their reasoning	• ☑ Differentiated instruction with leveled task cards (below/on/above grade level) • ☑ Manipulatives (fraction tiles) maintained for hands-on support • ☑ Graphic organizers (Keep-Change-Flip procedural anchor) • ☑ Flexible small-group reteaching using a different strategy number line instead of area model ☑ Parent contact with at-home digital resources like Khan Academy	• ☑ Independent Projects: Students create their own Fraction Division Story Book with at least three original real-world problems, visual fraction models, equations, and written explanations. According to the Michigan Department of Education (n.d.), from page 44, creating a story context for a division expression is an explicit expectation of the standard. Enriched students are challenged to create multiple contexts with increasing complexity. • ☑ Peer Tutoring / Group Leader: Advanced students serve as mathematical experts within cooperative groups, responsible for asking clarifying and probing questions of peers rather than supplying answers. As Dance and Kaplan (2018) explain

			from Chapter 5, student questioning enables students to advocate for their own learning and develop their own understanding. Enriched students practice the advanced skill of asking questions that assess understanding. • ☑ High-Level Centers (Desmos): Students explore what happens to the quotient as the divisor gets smaller, connecting fraction division to proportional reasoning introduced in the 6th grade ratio and proportion standards. • ☑ Games: This includes a Fraction War card game where students draw two fraction cards and race to determine which player's quotient is greatest, requiring mental estimation and written equation verification. • ☑ Use Vocabulary to Write Sentences: Students use fraction-specific academic vocabulary, quotient, divisor, dividend, reciprocal, visual model, in mathematical sentence stems to solidify conceptual understanding alongside procedural fluency.

<u>Common Core State Standard from page 44:</u> Apply and extend previous understandings of multiplication and division to divide fractions by fractions As the Michigan Department of Education (n.d.) explains on page 44, the standard for this lesson requires students to apply and extend previous understandings of multiplication and division to divide fractions by fractions. According to the Michigan Department of Education (n.d.), from page 44, students must be able to interpret and compute quotients of fractions and solve word problems involving division of fractions by fractions by using visual fraction models and equations to represent the problem. As the Michigan Department of Education (n.d.) further details on page 44, this standard is operationalized through tasks such as creating a story context for (2/3)÷(3/4) and using a visual fraction model to show the quotient. Furthermore, as the Michigan Department of Education (n.d.) states on page 44, students are expected to use the relationship between multiplication and division to explain that (2/3)÷(3/4)=8/9 because 3/4 of 8/9 is 2/3, applying the general formula (a/b)÷(c/d)=ad/bc to diverse real-world scenarios. By grounding the lesson in these technical requirements, the educator can then translate this complex academic language into a learning objective that is accessible to the students. <u>Learning Objective:</u> (In words that your students would understand) I can divide one fraction by another fraction by flipping the second	To support the mastery of this standard and to figure out if they learned it, the lesson will incorporate strategic questioning. As Dance and Kaplan (2018) suggest from Chapter 5, the Common Core State Standards for Mathematical Practice encourage students to provide evidence of their thinking and evaluate the evidence provided by others. This is facilitated through deliberate and direct questioning, both from the teacher and by the students themselves, as noted by Dance and Kaplan (2018) from Chapter 5.		

fraction and multiplying. I can draw a picture or write an equation to prove my answer makes sense, and I can create a real-life story that matches a fraction division problem. As the Michigan Department of Education (n.d.) explains on page 44, students should be able to show their work using pictures and equations to solve word problems, such as figuring out how many servings of yogurt fit in a partial cup or how to share a partial pound of chocolate equally. By translating the academic standard into accessible language, students can better understand what is expected of them. As Dance and Kaplan (2018) suggest from Chapter 5, the Common Core State Standards for Mathematical Practices should be designed to encourage students to provide evidence of their thinking and evaluate the evidence provided by others — this objective directly supports that goal.			

Materials:

- Fraction tiles and fraction bar manipulatives
- Dry-erase boards and markers (one per student)
- Real-world fraction division task cards like yogurt, chocolate, and land measurement scenarios from Michigan Department of Education (n.d.)
- Graph paper for visual fraction models
- Document camera / projector
- Number lines
- Student math journals
- Exit ticket slips

Questioning sentence stem anchor chart thought of from Dance and Kaplan (2018) in Chapter 5

Select your learning strategy:

- Direct Teach

- Demonstration
- Cooperative Learning
- Discovery/Inquiry-Based Learning
- Project-Based Learning
- Reading/Writing/Math Workshop
- Other

☑ Cooperative Learning + ☑ Discovery/Inquiry-Based Learning

Students work in small groups to investigate real-world fraction division scenarios using manipulatives and visual models before the algorithm is formalized. As Dance and Kaplan (2018) explain from Chapter 5, one of the best ways to help students provide evidence of their thinking is through deliberate and direct questioning. This cooperative structure creates natural opportunities for that questioning to occur, from both teacher and students. Thus, also supporting inquiry based learning as well.

Activities Planned: ☐Active (Students are active participants in learning)
☐Passive (Teacher led lecture/demonstration)
☐Both

☑ Both — Active and Passive

Teacher-modeled direct instruction, passive, paired with hands-on cooperative group investigation and student-led sharing, active.

Lesson Delivery Steps:

Step 1 — Launch / Pre-Assessment (10 minutes)

Display a real-world fraction scenario without numbers and use a I Notice / I Wonder routine. Ask the following: What do we know? What are we trying to figure out? As Dance and Kaplan (2018) suggest from Chapter 5, leaving out the numbers and encouraging students to notice and wonder through questioning creates a low-stress entry point that promotes deeper engagement with the problem. This also serves as a diagnostic pre-assessment of prior knowledge of fraction operations.

Step 2 — Explicit Teaching / Modeling (12 minutes)

Using the document camera, model (2/3) ÷ (3/4) step-by-step with a visual fraction model while thinking aloud about why flipping and multiplying works. According to the Michigan Department of Education (n.d.), from page 44, students must use the relationship between multiplication and division to explain that (2/3) ÷ (3/4) = 8/9 because 3/4 of 8/9 is 2/3. The general formula (a/b) ÷ (c/d) = ad/bc is introduced and linked to the visual model.

Step 3 — Guided Practice with Strategic Questioning (15 minutes)

Present two real-world word problems from the standard. Students use fraction tiles and graph paper to model solutions. Teacher circulates asking strategic questions. As Dance and Kaplan (2018) suggest from Chapter 5, guiding questions allow students to deepen their own reasoning as they justify their thinking and help navigate students through misconceptions. Example questions are as follows: What does the number mean in your equation? Why did you flip the second fraction? Can you use a second strategy to prove your thinking?

Step 4 — Cooperative Group Work (15 minutes)

Groups of 3–4 receive task card scenarios. Each student writes their own equation, draws a visual model, and writes one justification sentence. Groups discuss and compare strategies before the class share.

Step 5 — Class Share & Debrief (5 minutes)

Two or three groups share under the document camera. Teacher facilitates peer questioning using the sentence stem anchor chart. As Dance and Kaplan (2018) explain from Chapter 5, student questioning is just as important as teacher questioning because it enables students to advocate for their own learning and develop their own understanding.

Step 6 — Exit Ticket (3 minutes)

Students independently solve the following question: How many 3/4-cup servings are in 2/3 of a cup of yogurt? They then write one sentence explaining their reasoning. As the Michigan Department of Education (n.d.) establishes on page 44, this exact real-world context is embedded in the standard, making it an authentic and aligned summative check.

Check the Core Teaching Practices addressed in your lesson: (from MDE CTPs)

☑ **CTP #1 — Leading a group discussion:** Whole-class share and the I Notice / I Wonder launch routine provide structured opportunities for teacher-led and student-led discussion.

☑ **CTP #2 — Explaining and modeling content, practices, and strategies:** The explicit teaching phase uses think-aloud modeling of visual fraction models and the (a/b) ÷ (c/d) = ad/bc rule with the document camera.

☑ **CTP #3 — Eliciting and interpreting individual students' thinking:** Strategic teacher questioning during guided practice elicits individual student reasoning. As Dance and Kaplan (2018) explain from Chapter 5, teachers must think about which questions will allow them to formatively assess student understandings and/or misconceptions.

☑ **CTP #10 — Building respectful relationships with students:** The classroom culture emphasizes that questions show care for others' thinking. As Dance and Kaplan (2018) note from Chapter 5, we instill the belief in students that they learn the most by asking questions, and that it is important to ask when there is something you do not understand or are curious about.

☑ **CTP #15 — Check for Understanding:** Multiple checkpoints: the guided practice observation, student self-assessment during share, cooperative group task, and exit ticket.

List real-world connections including attention to English language learners and culturally and historically responsive practices (diversity, inclusion, equity, and social justice):

Real-World Connections:

All problems are grounded in tangible contexts drawn directly from the standard. According to the Michigan Department of Education (n.d.), from page 44, problems include sharing chocolate, measuring yogurt servings, and calculating land dimensions, which are scenarios students may encounter at home or in their communities.

English Language Learners:

Sentence stems and conversation stems are posted throughout the lesson. As Dance and Kaplan (2018) suggest from Chapter 5 through the persona of the chapter, question and conversation stems can be extremely helpful in getting students, especially English learners, to ask questions and discuss mathematical concepts. Key vocabulary, such as quotient, fraction model, and equation, is pre-taught with visual supports, and problems are read aloud paired with diagrams.

Culturally and Historically Responsive Practices:

Task card scenarios include culturally diverse food and community contexts so all students can see themselves in the mathematics. Students are also invited to create their own fraction division story problems rooted in their own lived experiences, positioning their cultural knowledge as mathematically valid.

List technology tools:

- Document camera for teacher modeling (Step 2) and student sharing (Step 5)
- Projector / interactive whiteboard for displaying problems and recording class strategies on the math wall
- Desmos Fraction Tool optional for enrichment via students who finish early to explore visual models digitally
- Khan Academy for differentiated at-home review of fraction division skills

List collaboration opportunities (whole group, small group, partnerships, building resource personnel i.e., school social worker, special educators, parents, etc.):

Whole Group: I Notice / I Wonder launch, teacher-modeled instruction, class share and debrief with peer questioning using sentence stems.

Small Group: Cooperative task card activity, groups of 3–4. Each student is individually accountable for their equation and model but discusses reasoning with their group.

Building Resource Personnel: Special educator / co-teacher supports students with IEPs during guided practice. School math interventionist may pull a small Tier II group during Step 4 if pre-assessment data warrants it.

Lesson Plan Guide

Assessment: Used to gather information about a student's progress towards mastery of the learning objective, help the teacher identify what instruction is working well and what needs refinement, and informs the students about their learning.

Options to consider

☑ Diagnostic/Pre-Assessment:

The I Notice / I Wonder launch serves as a pre-assessment. Teacher observes which students connect the scenario to multiplication and division of fractions and which students need additional scaffolding to access the problem context. No written product is required, and the teacher notes observations.

☑ Formative — During the Lesson:

Teacher circulates during guided practice and cooperative group work using the three strategic question types identified by Dance and Kaplan (2018) from Chapter 5: Questions that clarify and probe for justification, questions that guide, challenge, and extend thinking, and questions that assess understanding. Think/Pair/Share is used during Step 3. Students also complete a quick thumb self-assessment before the class share to give the teacher an immediate snapshot of class-wide confidence.

☑ Summative — Exit Ticket:

Students independently solve one fraction division word problem and write one sentence explaining their reasoning. As the Michigan Department of Education (n.d.) establishes on page 44, the exit ticket problem how many 3/4-cup servings are in 2/3 of a cup of yogurt is taken directly from the standard, making it an authentic and aligned measure of student mastery.

Intervention: How will we respond when they don't learn?

- ☑ Differentiated Instruction — Leveled task cards (below, on level, above) allow the teacher to assign appropriately challenging problems while keeping the whole class engaged in the same concept.
- ☑ Manipulatives — Fraction tiles remain available throughout and during intervention for students who need concrete support alongside abstract symbolic work.
- ☑ Graphic Organizers — A visual Keep-Change-Flip step-by-step organizer provides a procedural anchor for students who struggle with abstract manipulation.
- ☑ Immediate Feedback via Questioning — As Dance and Kaplan (2018) describe from Chapter 5, guiding questions during work time help navigate students through misconceptions as they explain their reasoning, using questions such as "What does the number mean in your equation?" or "Would it help to draw a picture?"
- ☑ Flexible Grouping — Students who need support are grouped with the teacher or co-teacher during the cooperative task for a mini re-teach using a story-based context before symbolic notation.

☐Differentiated Instruction
☐Target specific skills
☐Data item analysis
☐Leveled materials (below, on level, above)
☐Bloom's Taxonomy
☐Grade recovery (re-do/correct)
☐Parent contact

☐Referral to Student Support Team
☐Graphic organizers
☐Manipulatives
☐Choice boards
☐Immediate feedback
☐Flexible grouping
☐Extended responses (math/reading)
☐Journal/Reading logs

Responses to Intervention (RtI)

- ☑ Small Group Instruction (Tier II) — Teacher or interventionist pulls a small group the following class period for targeted reteaching using simpler whole-number division to rebuild the conceptual foundation.
- ☑ Re-teach in a Different Way — Reteaching uses a different entry point (e.g., number line models instead of area models). As Dance and Kaplan (2018) note from Chapter 5, students benefit when instruction is modified by backtracking to build background knowledge through a new approach.
- ☑ Tutoring — Before or after school sessions for students who remain below mastery after Tier II support.
- ☑ Referral to Student Support Team — If student continues to struggle after Tier I and II supports, a referral to the Student Support Team is initiated for comprehensive review.

☐Small group instruction
☐Tiered group instruction (Tier I, II, III)
☐1-1
☐Centers (leveled)
☐Re-teach in a different way
☐Modify: backtrack, build background knowledge
☐Tutoring: after or before school, lunch
☐Referral to Student Support Team

Enrichment: How will we respond if they already know it?

- ☑ Independent Projects — Students create their own "Fraction Division Story Book" with at least three original real-world problems, visual fraction models, equations, and written explanations. According to the Michigan Department of Education (n.d.), from page 44, creating a story context for a division expression is an explicit expectation of the standard — enriched students are challenged to create multiple contexts with increasing complexity.
- ☑ Peer Tutoring / Group Leader — Advanced students serve as mathematical experts within cooperative groups, responsible for asking clarifying and probing questions of peers rather than supplying answers. As Dance and Kaplan (2018) explain from Chapter 5, student questioning enables students to advocate for their own learning and develop their own understanding — enriched students practice the advanced skill of asking questions that assess understanding.
- ☑ High-Level Centers (Desmos) — Students explore what happens to the quotient as the divisor gets smaller, connecting fraction division to proportional reasoning introduced in the 6th grade ratio and proportion standards.
- ☑ Games — "Fraction War" card game where students draw two fraction cards and race to determine which player's quotient is greatest, requiring mental estimation and written equation verification.
- ☑ Use Vocabulary to Write Sentences — Students use fraction-specific academic vocabulary (quotient, divisor, dividend, reciprocal, visual model) in mathematical sentence stems to solidify conceptual understanding alongside procedural fluency.

☐Choice boards
☐Use vocabulary to write sentences
☐Accelerated reader
☐Centers-High level
☐Reading buddies
☐Peer tutoring

☐Enriched-Leveled Reader-Novels
☐Picture/writing journals
☐Independent projects
☐Separate curriculum
☐Games
☐Group leader